

# Chapter 14

## Applications Of Inversive Geometry

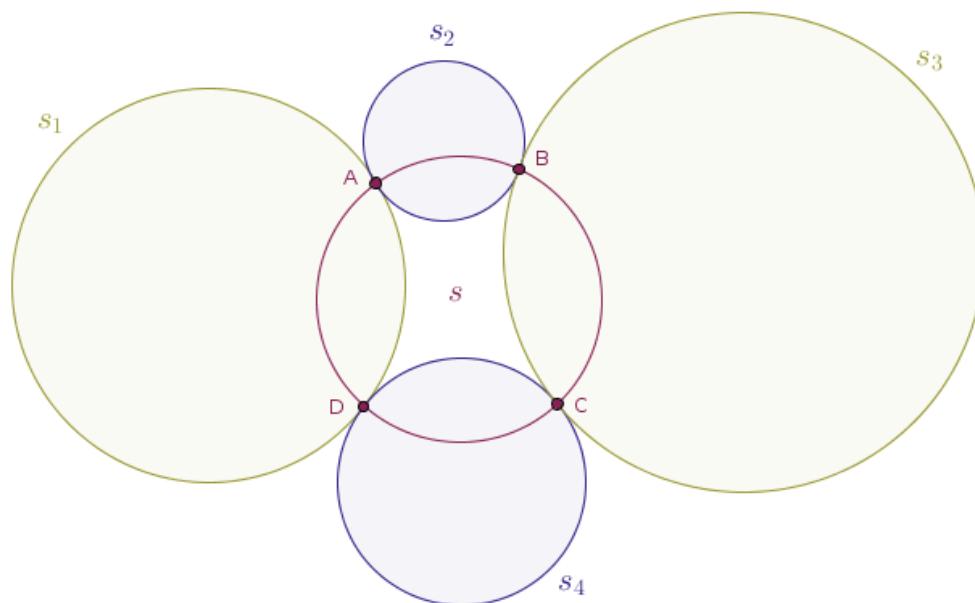
### Four Pairwise Externally Tangent Circles

Starting from this chapter onward we will look at how the tiny bit of theory that we have developed so far can be applied in various mathematical settings in two different directions. First, we will apply inversion of the real plane in a circle to solving problems which, at first, blush have nothing to do with inversive geometry. Second, we will make inversion of the real plane in a circle a small building block of yet another theory, the theory of pencils of circles.

To that end, in the Problem 03 and the Problems 04 we have already seen that a skillful application of inversion of the real plane with respect to a circle makes certain purely computational tasks almost a breeze.

In the next problem we will investigate how a tasteful application of inversion of the real plane with respect to a circle can make the generation of a proof very straightforward if not almost mechanical.

**Problem 20 (four pairwise externally tangent circles).** Let four fixed distinct coplanar circles  $s_1, s_2, s_3, s_4$  touch externally pairwise at the points  $A, B, C$  and  $D$  correspondingly (Figure 14.1):



**Prove:** that the said tangency points  $A, B, C$  and  $D$  of the said circles  $s_1, s_2, s_3, s_4$  belong to the same circle,  $s$ . Alternatively, prove that the quadrilateral  $ABCD$  is cyclic.

**Solution.** Before we generate the requested proof, we would like to observe in passing that our original objective of proving that the said tangency points  $A, B, C$  and  $D$  belong to the same circle,  $s$ , may be somewhat revealing. Presented in the context of inversive geometry, when the related problems come at us hard and fast, it seems only natural and automatic to try pouring the inversive geometry oil on such a problem.

In order to remove the idea or the hint of inversion from the immediate surface of the problem, we can give a slightly different but, of course, an equivalent formulation of the objective based on a cyclic quadrilateral.

The point of the last maneuver is, in a mathematical competition or a mathematical Olympiad setting or in vacuum, it may not be at all that obvious the machinery of which mathematical layer will be able to grind a given problem into a fine powder the fastest. The people who design the problems for mathematical competitions are very good at very cleverly hiding or camouflaging the said machinery from plain sight, making it very easy for a non-savvy problem-solver to go in the wrong direction.

As such, now that we know about this particular transformation of the real plane, we will not be shy about explicitly rephrasing the problem statement in a more suggestive way so that it becomes painfully

clear that we are about to unleash inversion with respect to a circle on a non-suspecting problem at hand.

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We would like to demonstrate this idea on a problem that is not too difficult and point out the fact that the objective of all the problems in this text is not to find a solution of a problem that is necessarily *simpler*, in some sense, but to find a solution that employs specifically *inversion*. Such an inversion-based solution may or may not turn out to be “simpler” than its alternative counterparts. Comparing various inversion-based solutions to their alternatives is a very useful exercise in its own right.

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Oftentimes the actual magnitude of the length of the radius of inversion in a particular setting is absolutely irrelevant. Yes, in our introductory text we pedantically dragged the said radius of inversion all over the place, but there was a good academic reason for that. The fastest and the easiest way to demonstrate the above irrelevancy point is to make the students witness it for themselves.

Because in this particular problem all the tangency points are created equal, we will choose the point  $A$  to be the center of the inversion of the parent plane with positive power but our readers are more than welcome to experiment with any other remaining tangency point on their own.

Moreover, we will not even bother with specifying the length  $r$  of the radius of the said inversion other than stating that it, of course, is greater than zero.

The angle between the tangent straight lines drawn to the input circles  $s_{1,2}$  through the point of their tangency  $A$  is either zero or any integral multiple of  $\pi$ . Either way, these two tangent straight lines, for all practical purposes, are parallel. Hence, the circles  $s_{1,2}$  are parallel also (sounds funny, doesn't it?).

Therefore, under the inversion of the parent plane centered at the point  $A$  with positive power the circles  $s_{1,2}$  will be taken into two straight lines  $s'_{1,2}$  that will be parallel also (see the Theorem 35 and the Case 1 of the Theorem 33).

By the same Theorem 33, the remaining circles  $s_{3,4}$  will be transformed into the circles  $s'_{3,4}$  that will touch:

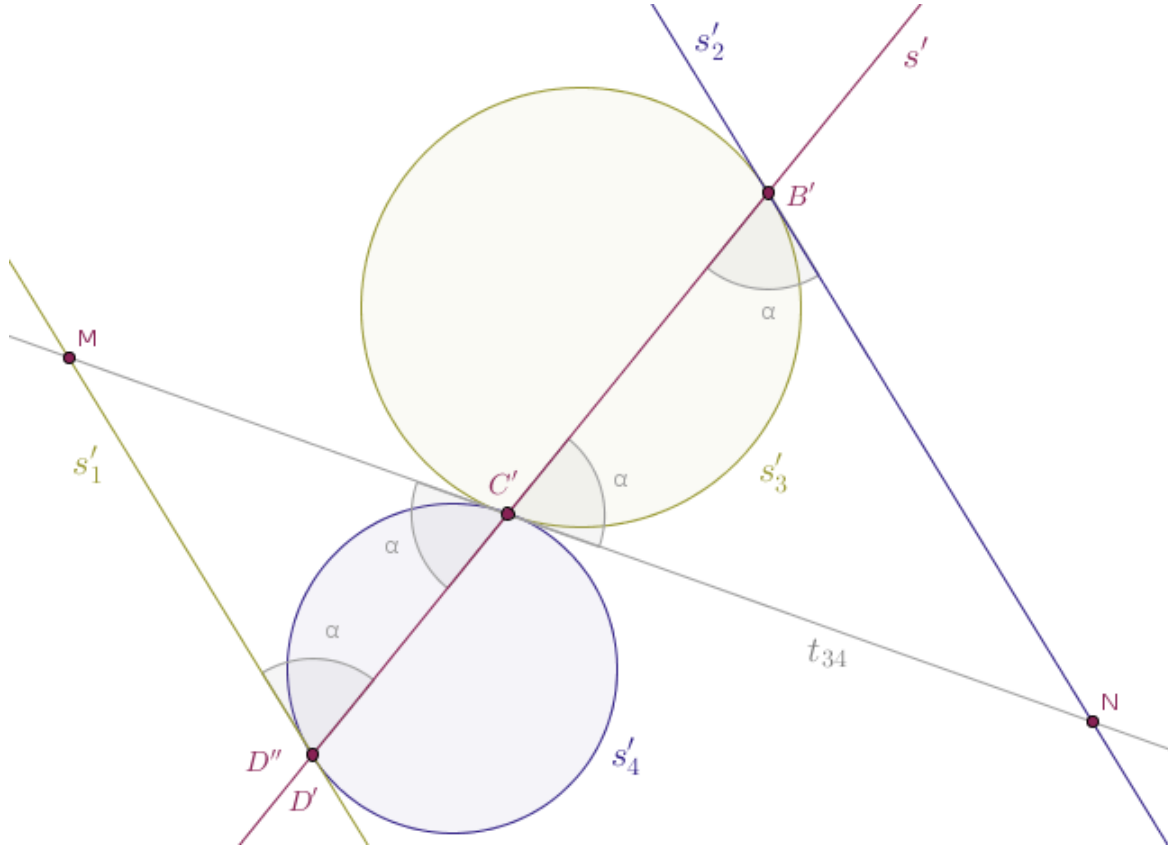
- each other and
- the parallel straight lines  $s'_{1,2}$

By the Theorem T15, the circle  $s$  will be transformed into a straight line  $s'$ .

Let the gray straight line  $t_{3,4}$ , that is a common internal tangent of the circles  $s'_{3,4}$  at the point of their tangency  $C'$ , intersect\* the straight lines  $s'_{1,2}$  at the points  $M$  and  $N$  correspondingly.

Let  $B'$  be:

- the point of tangency of the objects  $s'_2$  and  $s'_3$  and
- the point where the straight lines  $s'_2$  and  $s'$  intersect (Figure 14.2):



Lastly, let us pretend that the point that currently has two labels of  $D'$  and  $D''$  amounts to two *distinct* points:

- let  $D'$  be the point where the objects  $s'_1$  and  $s'_4$  touch and
- let  $D''$  be the point where the straight lines  $s'_1$  and  $s'$  intersect

Clearly, if we succeed in proving that the points  $B'$ ,  $C'$ ,  $D'$  belong to the same straight line  $s'$  and that the points  $D'$  and  $D''$  are indistinguishable then we will, effectively, prove that the corresponding points  $A$ ,  $B$ ,  $C$  and  $D$  belong to the same circle  $s$ .

The point  $N$  that simultaneously belongs to two straight lines tangent to the circle  $s'_3$  is equally distanced from the tangency points  $B'$  and  $C'$ :

$$|NB'| = |NC'|$$

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Therefore, the triangle  $B'NC'$  is isosceles and, by the famous “Pons Asinorum” or [Euclid’s “Elements” Book 1 Proposition 5](#), the interior angles at the base of the said triangle, marked as  $\alpha$ , must be equal one another:

$$\angle NB'C' = \angle NC'B' = \alpha$$

Moreover, the corresponding angles at the point  $C'$  are vertical and, by [the same Book 1 Proposition 15](#), are also equal one another:

$$\angle NC'B' = \angle MC'D'' = \alpha$$

Then on the one hand, because the straight line  $s'$  intersects two distinct straight parallel lines  $s'_{1,2}$ , by [Euclid’s “Elements” Book 1 Proposition 29](#), the corresponding *alternate* angles are also equal one another:

$$\angle MD''C' = \angle NB'C' = \alpha$$

implying that it is the case that:

$$\angle MD''C' = \angle MC'D''$$

and by the converse of “Pons Asinorum” or [Euclid’s “Elements” Book 1 Proposition 6](#), if the above relation holds, which it does, the corresponding triangle  $C'MD''$  must be isosceles:

$$|MD''| = |MC'|$$

But on the other hand, by the same tangents-of-equal-length argument, the point  $M$  that simultaneously belongs to two straight lines tangent to the circle  $s'_4$  is equally distanced from the tangency points  $D'$  and  $C'$ :

$$|MD'| = |MC'|$$

Therefore, from the last two relations it follows that:

$$|MD'| = |MC'| = |MD''|$$

meaning that:

$$|MD'| = |MD''|$$

implying that the points  $D'$  and  $D''$  are, in fact, one and the same:  $D' \equiv D''$ .

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Therefore, the straight line  $s'$  intersects the object  $s'_1$  at the same point where the objects  $s'_1$  and  $s'_4$  touch. Therefore, the points  $B'$ ,  $C'$  and  $D'$  belong to the same straight line  $s'$  and, therefore, the tangency points  $A$ ,  $B$ ,  $C$  and  $D$  belong to the same circle,  $s$ .  $\square$

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\*The said common internal tangent,  $t_{34}$ , of the image circles  $s'_{3,4}$  at the point of their tangency,  $C'$ , has only two choices for its mutual juxtaposition relative to the parallel straight lines  $s'_{1,2}$ :

- that tangent straight line is either parallel to  $s'_{1,2}$  or
- not parallel to  $s'_{1,2}$

If the tangent  $t_{34}$  is not parallel to  $s'_{1,2}$  then it will intersect these parallel straight lines as was explained above.

If the tangent  $t_{34}$  **is** parallel to  $s'_{1,2}$  then the straight line  $s'$  will coincide with the straight line of centers of the image circles  $s'_{3,4}$ , which would imply that in this case there is no need to assume that the points  $D'$  and  $D''$  are distinct.

Indeed, in this case the straight line of centers of the image circles  $s'_{3,4}$  will be perpendicular to the parallel lines  $s'_{1,2}$  and will, thus, house the tangency points  $B'$ ,  $C'$  and  $D'$ .